Juan Martin N. Rodrigo

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EDUCATION

University of the Philippines Diliman

MS Physics *July 2026 (expected)*
Thesis adviser: Professor Ian Vega Current GPA: 1.06 / 1.00*
Working thesis title: *Induced metrics in black hole thermodynamic geometry*

BS Physics, *summa cum laude* *July 2024*
Thesis adviser: Professor Ian Vega GPA: 1.15 / 1.00*
Thesis title: *On the uniqueness of flat thermodynamic geometries*

** In the UP system, 1.00 is the highest possible grade.*

EMPLOYMENT

Instructor *August 2024 – present*
National Institute of Physics, UP Diliman

RESEARCH INTERESTS

General relativity, thermodynamic geometry. Investigating fundamental geometric structures in thermodynamic state spaces and their applications to self-gravitating systems and black holes.

PUBLICATIONS

1. **J. Rodrigo** and I. Vega (2025). Interacting systems with zero thermodynamic curvature. *Physical Review E,* 111(5), 054120. doi.org/10.1103/PhysRevE.111.054120

Conference proceedings

1. **J. Rodrigo** and I. Vega, *On the uniqueness of flat thermodynamic geometries,* Proceedings of the 41st Samahang Pisika ng Pilipinas Physics Conference, July 2023.
2. **J. Rodrigo**, J. L. Villarin, L. Geraldez, A. Mendoza, and I. Vega, *Dynamics of Newtonian and relativistic thin fluid shells,* Proceedings of the 39th Samahang Pisika ng Pilipinas Physics Conference, October 2021.

HONORS AND AWARDS

DOST Merit Scholarship *2019 – 2023*
Merit scholarship (~Php 90,000 per year + tuition) for undergraduate studies in STEM.

University Scholar *2019 – 2023*
Awarded for GPA < 1.45 in a semester.

TEACHING EXPERIENCE

Instructor on record, University of the Philippines Diliman
Typical class size of 30 (18) students for discussion (lab) sections; 9 contact hours / week.

- **Physics 71 (71.1)**, Elementary Physics I (lab) *1st term AY 2024-25, 1st term AY 2025-26*
  Calculus-based class on Newtonian mechanics, fluids, and waves.
- **Physics 72 (72.1)**, Elementary Physics II (lab) *2nd term AY 2024-25*
  Electromagnetism and optics.
- **Physics 73**, Elementary Physics III *1st term AY 2025-26*
  Thermodynamics, special relativity, and quantum mechanics.

Grader, University of the Philippines Diliman

- **Physics 118**, Mathematical Physics III *1st term AY 2024-25*
  Math methods for physics majors; topics include: group theory, differential geometry.
- **Physics 151**, Statistical Physics I *2nd term AY 2024-25*
  Thermodynamics and statistical physics for physics majors.

SKILLS

Programming skills: Mathematica and Python (proficient); C/C++, Javascript, Julia, and Bash (basic); LaTeX, HTML, CSS, Markdown.

Languages spoken: English (native), Tagalog (fluent).

REFERENCES

Available upon request.